



Predict, Then Run

Quick facilitation notes & answer key

Goal

Students read a program, predict the landing number before running, then run to confirm — and alter a block and predict again, describing the effect. Reading and predicting code outcomes, and describing how a change affects the result, is the heart of C3.2.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 to check predictions by walking the code.

Run it (about 20 min)

1. I DO: read Code A aloud without moving — 'start 0, forward 2 would be 2, forward 2 would be 4 — I predict 4.' Then run it to confirm.
2. WE DO: Exercise 1 — predict (4) then run Code A. Exercise 2 — predict Code B (2), then run.
3. YOU DO: Exercise 3 — change back 2 to back 1; predict (3) and run; describe that a smaller back means Bit lands one higher.
4. Connect: 'Reading and predicting first is how coders check their work before running it.'

Answer key (modelled by the run-gate)

Ex 1 — Code A: 0 → forward 2 → 2 → forward 2 → 4. Predict and land on 4.

Ex 2 — Code B: 5 → back 1 → 4 → back 2 → 2. Land on 2.

Ex 3 — change back 2 to back 1: 5 → back 1 → 4 → back 1 → 3. Bit lands on 3 — one higher than before, because the last block now takes away one fewer step.

Why predicting works: the blocks run in a fixed order, so you can trace them in your head — start number, then each move — to know the result before running.

Watch for & differentiate

Common slip: predicting the final number by adding all the move numbers and ignoring back blocks — remind students that back takes away.

Simplify: trace each code together on the floor path, pausing to predict the next number block by block.

Extend: ask students to predict Code A if the start block were 'start at 1' (lands on 5, the end).

Success indicator: the child writes a correct prediction before running and confirms it, and explains the effect of the changed block.